

Poojari Venkata Shreesh

AI/ML Engineer

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SUMMARY

Built production-grade AI pipelines with LLM orchestration, RAG, and workflow validation for travel planning and document digitization. Experienced in integrating Gemini, FastAPI, Next.js, Neo4j, and external APIs to ship end-to-end systems with reliable backend orchestration and polished frontend UX.

EDUCATION

Arizona State University

Master of Science in Computer Science (MScS) | GPA: 3.93

Relevant Coursework: Artificial Intelligence, Knowledge Representation and Reasoning, Machine Learning, Data Processing at Scale

Tempe, AZ

Aug 2024 – Present

BITS Pilani

Bachelor of Technology in Computer Science | GPA: 7/10

Hyderabad, India

Jul 2018 – May 2022

SKILLS

Languages: Python, C++, TypeScript, C, Java

AI/ML & LLM: LLM Orchestration, Gemini API, Prompt Engineering, RAG Systems, Graph Embeddings, Entity Extraction, Semantic Chunking

Knowledge Representation: Neo4j, Knowledge Graphs, Logic Programming (ASP/Clingo), Constraint Satisfaction, Graph Algorithms

ML Frameworks & Tools: Scikit-learn, NLTK, TensorFlow, PyTorch, Pydantic, Ollama, LLaVA

Backend & Systems: FastAPI, Workflow Orchestration, Multi-Agent Systems, Session Management, Caching, API Design, REST Frontend: Next.js, React, TypeScript, Zustand, Tailwind CSS, Authentication (Auth0)

Databases & APIs: Neo4j, MongoDB, PostgreSQL, Google Places API, Google Routes API, External API Integration

DevOps & Tools: Git, Linux, Bash/Zsh, Jupyter Notebooks, Uvicorn, Docker, CI/CD

Technical Domains: Systems Design, Software Architecture, Production Reliability, Computer Vision, NLP

PROFESSIONAL EXPERIENCE

Darwinbox

Software Development Engineer

- Collaborated as part of the core/platform team and contributed towards two projects.
- Designed and developed features for workflow pages to include tagging and download options, improving usability.
- Developed dynamic data cards, implemented using MongoDB.
- Modularized legacy workflow modules to personalize data on a user level instead of a team level.

Hyderabad, India

Jul 2022 – Jul 2023

BlueJeans by Verizon

Frontend Intern

- Collaborated on white box unit testing in the WMC team, resulting in a 5% increase in testing coverage.
- Developed unit test cases for the Web Meeting Client, increasing code testing coverage.
- Used Jest testing library to generate unit tests for ReactJS files and tracked progress using Git.

Bangalore, India

Jul 2021 – Dec 2021

BHEL

Web Development Intern

- Designed web pages for the company intranet to manage and save tenders for auctions.
- Developed web pages using PHP, CSS, and HTML.

Visakhapatnam, India

May 2020 – Jun 2020

PROJECTS

travelMate: AI-Assisted Travel Planning Platform

FastAPI, Python, Gemini LLM, Google Places/Routes APIs, Next.js, TypeScript, Workflow Orchestration

- Engineered a full-stack travel planning service that orchestrates LLM extraction, rule-based validation gates, and multi-stage workflow logic to transform natural-language trip requests into structured, route-aware itineraries.
- Designed and implemented sequential completeness and feasibility evaluators in FastAPI using Gemini to extract planning state, preventing low-quality or infeasible plan generation before expensive API calls.
- Integrated Google Places and Google Routes APIs with itinerary assembly and budget estimation pipeline; built session memory and response caching to support conversational continuity and reduce redundant computation.
- Developed Next.js frontend with chat-driven UX, Auth0 authentication with preference persistence, and real-time itinerary rendering; connected via robust API contract layers and proxy routing for seamless frontend-backend communication.

Accio: PDF/OCR Processing for Workflow Digitization

Neo4j, RAG, LLM, Knowledge Graphs, Python, NLP

- Developed a workflow digitization system that transforms PDF documents into structured Neo4j-backed knowledge graphs enriched with embeddings and exposed through a RAG interface.
- Integrated layout-aware OCR, semantic chunking, LLM triplet extraction, graph construction, and hybrid retrieval for grounded answer generation.
- Evaluated on 24 U.S. federal documents, populating a graph with 3,239 nodes and 4,941 relationships, achieving $F1 = 77.5\%$ for entity extraction and $NDCG@5 = 0.76-0.82$ for retrieval accuracy.

AI Fashion Stylist App

Google Gemini API, Ollama, LLaVA, Computer Vision, Node.js, AI Chat

- Built an AI-powered fashion stylist MVP with Node.js/Express, integrating Google Gemini and local Ollama (Llama 3, LLaVA) for chat and image analysis with a unified service layer.
- Implemented multi-image clothing upload, AI-powered tagging (type, color, style), and occasion-based outfit recommendations using the user's wardrobe context.
- Designed REST API for wardrobe CRUD, AI chat, recommendations, and image analysis; added JWT auth, rate limiting, Helmet, and CORS for security and cross-origin support.